Seminar6

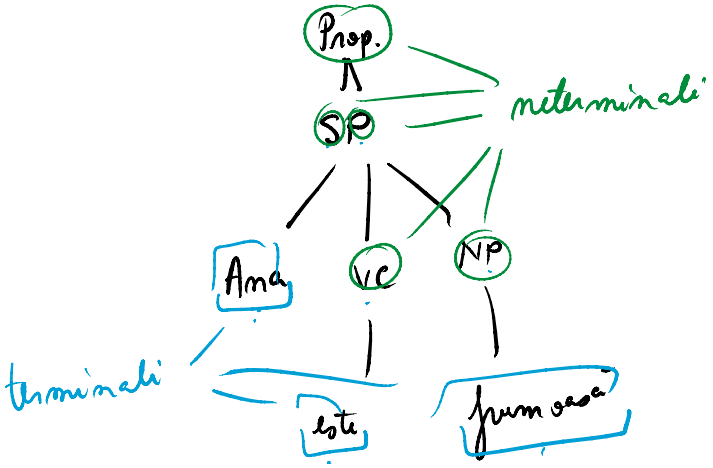
luni, 15 noiembrie 2021 10:03

Limbaje independente de context — LIC

Gramatici independente de context

Gramatica este un set de reguli de combinare a cuvintelor din vocabular pt. a forma propoziții în lb. română.

reguli → reguli
cuvinte → simboluri
vocabular → alfabetul
propoziții → șiruri
limba → limbajul



Prop → S P
S → Ana
P → VC NP
VC → este
NP → frumoasă

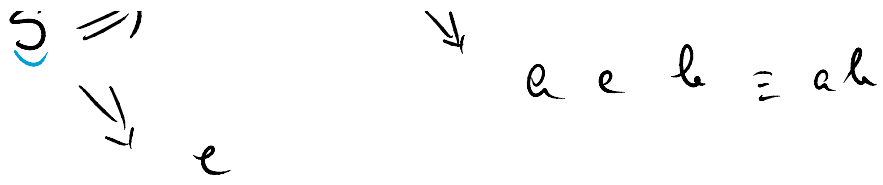
$S \to S_1 S_2$

$S \to a$

$S \to aSb \mid \varepsilon$ — $a^n b^n$

$S \Rightarrow aSb \Rightarrow aaSbb$

$aaebb = aabb$



$$G = (V, \Sigma, R, S)$$

- V: mulțimea de terminali și neterminali
- Σ: alfabetul / mulțimea de terminali
- R: mulțimea de reguli
- S: simbolul de start

$$R \subset (V \setminus \Sigma) \times V^*$$

- $(V \setminus \Sigma)$: neterminal
- V^*: concatenare de terminali și neterminali

$$\Sigma = \{0, 1\} \qquad V = \{A, B, C, 0, 1\}$$

$$R \begin{cases} A \to 0A \\ B \to ACB \\ C \to 01. \end{cases}$$

$S = A$.

~~e~~

~~aSb U e~~

$\cap$

$\times$

1. $S \to aSb$
2. $S \to e$

$\Rightarrow S \to aSb \mid e$

$\mathcal{L}(G)$

$n \cdot m$

$\mathcal{L}(G)$

G | $S \to aSb \mid e$ |

$a^n b^n$ $\quad \Sigma = \{a, b\}$

$L = \{ w \in \Sigma^* / w \text{ e de forma } a^n b^n, n \geq 0 \}$

$S \Rightarrow e \in \mathcal{L}(G)$

$S \Rightarrow aSb \Rightarrow aeb = ab \in \mathcal{L}(G)$

$\in \mathcal{L}(G)$

$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaaSbbb = aaabbb$

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1. $\Sigma = \{a, b\}$

$L = \{ w \in \Sigma^* / w \text{ e de forma } a^n b^m, n \geq m, n, m \geq 0 \}$

$\to S \to aS \mid aSb \mid e$

$a^{n-m} a^m b^m$

$S \Rightarrow aS \Rightarrow aaS \Rightarrow aaaSb \Rightarrow aaaaSbb \Rightarrow aaaabbb$

$S \to aSb \Rightarrow aaSb$

$S \to S_1 S_2 \mid e$

$S_1 \to aS_1 \mid e$

a^{n-m} (S_1) $a^m b^m$ (S_2)

$S_2 \to a S_2 b \mid e$

$S \Rightarrow S_1 S_2 \Rightarrow a S_1 \; a S_2 b$

$\Rightarrow aa S_1 \; aa S_2 bb$

$\Rightarrow aa \; aa bb$

2. $\Sigma = \{0, 1\}$

$L = \{ w \in \Sigma^* / w$ e de forma $(00)^k 1^k$, $k \geq 2 \}$

$S \to aSb \mid ab$ $\quad a^n b^n$, $n \geq 1$.

$S \Rightarrow aSb \Rightarrow aaSbb \ldots$

$S \to 00S1 \mid 0000 11$

$(00)^2 1^2$.

$(00)^k 1^k \quad k \geq 2$.

3. $\Sigma = \{0, 1\}$

$L = \{ w \in \Sigma^* / \#_0(w) : 2$ si $\#_w \not: 2 \}$

$S \to 00 \mid 1$ $\quad\quad\quad

1.

3. $\Sigma = \{0, 1, 2\}$

$L = \{ |x| \in \Sigma^* / \; |x| \text{ e de forma } 0^n 1^{n+m} 2^m \}$

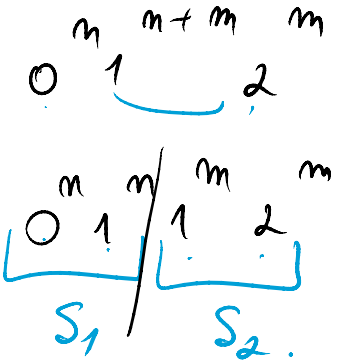
$S \to 01S \mid 12S \mid e$

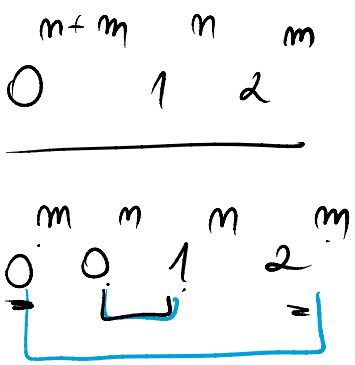
$S \Rightarrow 01S \Rightarrow 0101S$

$S \to S_1 S_2$

$S_1 \to 0S_1 1 \mid e$

$S_2 \to 1S_2 2 \mid e$



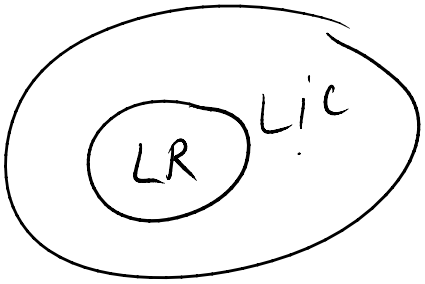


$S \to 0S2 \mid A$

$A \to 0A1 \mid e$



$S \Rightarrow 0S2 \Rightarrow 0A2 \Rightarrow 00A12$



$\forall L \in LR \Rightarrow L \in LIC$

$\forall L \in LIC \Rightarrow$ GIC.

$L \in LR$

$E = 0^*((1 \cup 2)(1 \cup 2))^*$

000 1221

G

$S \to ASA \mid 0S$

$A \to 1A \mid A1 \mid 2A \mid A2 \mid e$

$S \Rightarrow ASA \Rightarrow 1A0S2A$

$0^*((1 \cup 2)(1 \cup 2))^*$

$S_1 \quad S_2$

G:

$S \to S_1 S_2$

$S_1 \to 0S_1 \mid e$

$S_2 \to 1S_21 \mid 1S_22 \mid 2S_21 \mid 2S_22 \mid e$

5. $\Sigma = \{ a, b \}$

$L = \{ w \in \Sigma^* / w$ începe și se termină cu același simbol sau $\#_a(w) = \#_b(w) \}$

$S \to aS_2a \mid bS_2b \mid aS_1b \mid bS_1a$

$S_1 \to abS_1 \mid baS_1 \mid aS_1b \mid bS_1a \mid S_1ab \mid S_1ba \mid e$

$S_2 \to aS_2 \mid bS_2$

$S \Rightarrow aS_1a \Rightarrow aabS_1a \Rightarrow aaba$

aba

aba.

$S \to S_1 \mid S_2$.

$S_1 \to aS_3a \mid bS_3b \mid a \mid b$

$S_3 \to aS_3 \mid bS_3 \mid e$

$S_2 \to aS_2b \mid bS_2a \mid e$

6. $\Sigma = \{ a, b, c \}$

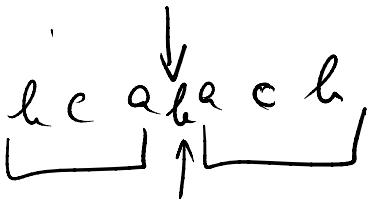
$L = \{ w \in \Sigma^* / w$ e un palindrom de lungime impară $\}$.

$L = \{ w \in \{a,b,c\}^* \mid$ de lungime impară $\}$.

$S \to aSa \mid bSb \mid cSc \mid a \mid b \mid c$

$S \Rightarrow bSb \Rightarrow bcScb$

$\Rightarrow bcaSacb$



$\Sigma = \{0, 1, 2\}$

$L = \{ w \in \Sigma^* \mid w$ e de forma

$0^p 1^k 2^r$, $k > r$,

$p \geq 2$,

$k, r \geq 0 \}$.

$k \geq 1$.